

GraphCompose

v2.1.0 *twinned*

Open-source Java library for generating structured business PDF documents with a declarative DSL.

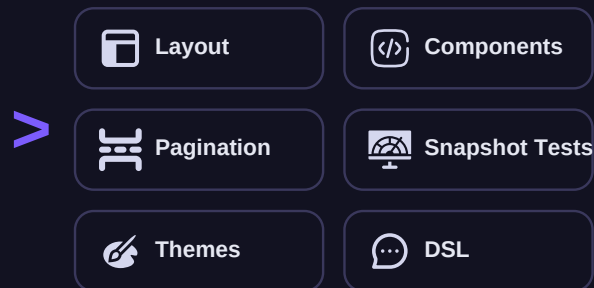
1 Java · Declarative DSL

```
var out = Path.of("deck.pdf");
try (var doc = GraphCompose
    .document(out)
    .pageSize(DocumentPageSize.A4)
    .create()) {
    doc.pageFlow(page -> page
        .addParagraph("GraphCompose")
        .addParagraph("Cinematic."));
    doc.buildPdf();
}
```

You describe the document intent — the engine resolves the rest.

2 GraphCompose Engine

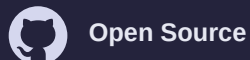
Semantic graph · deterministic layout



3 Output Backends

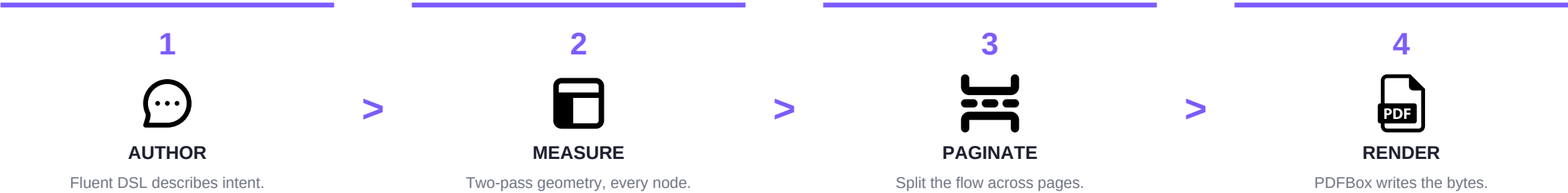


4 Real PDF Output



From one Java file to a designed PDF

You describe the document semantically — **sections, rows, tables, charts, shapes, layers** — and the engine resolves the rest: it **measures** every node twice, **paginates** the flow row-by-row, and **renders** through an isolated PDFBox backend. No manual coordinates, no XML templates.



Deterministic

The same input renders the same bytes on every machine — layout is reproducible, not best-effort.

Regression-tested

layoutSnapshot() diffs the geometry and PdfVisualRegression pixel-tests fonts and colour, right in your pull requests.

Production-ready

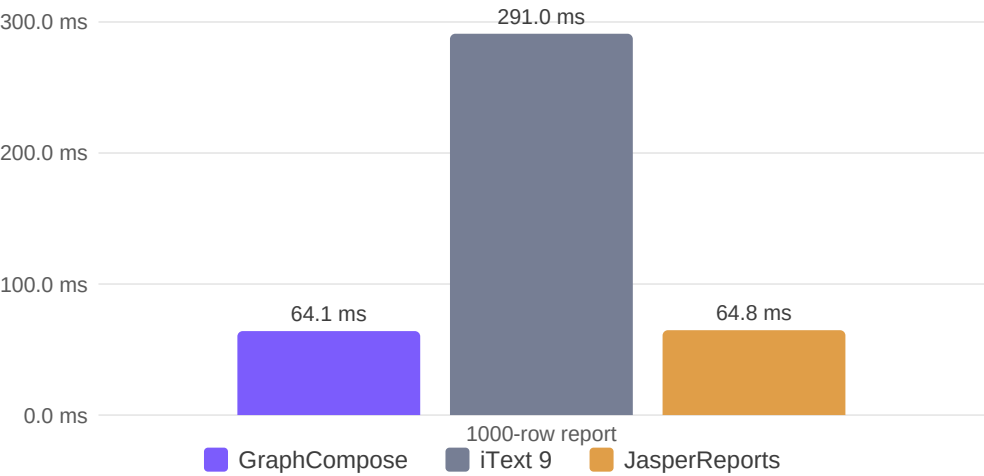
An isolated PDFBox backend streams straight to an HTTP response — no temp files, no manual coordinates.

GraphCompose vs. the field

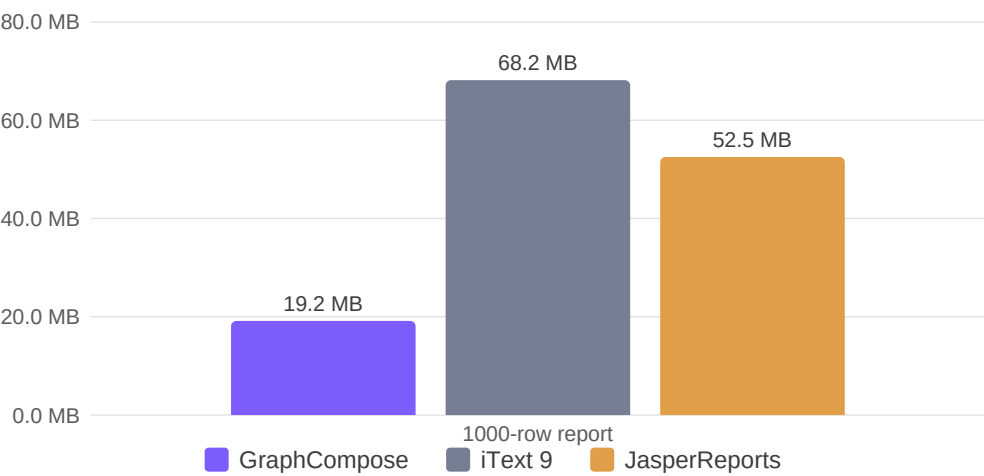
The same documents through three engines — render time and peak heap, measured by this repository's own harness. The table and charts below are drawn by GraphCompose straight from the result file.

Report size	GraphCompose	iText 9	JasperReports
1 page · 3 lines	2.8 ms · 0.2 MB	2.8 ms · 0.2 MB	5.7 ms · 0.2 MB
40 rows	5.7 ms · 1.0 MB	14.5 ms · 3.1 MB	9.9 ms · 2.5 MB
200 rows	15.3 ms · 3.9 MB	57.6 ms · 13.8 MB	21.1 ms · 10.8 MB
1000 rows	64.1 ms · 19.2 MB	291.0 ms · 68.2 MB	64.8 ms · 52.5 MB

Render time at 1000 rows — ms (lower is faster)



Peak heap at 1000 rows — MB (lower is lighter)

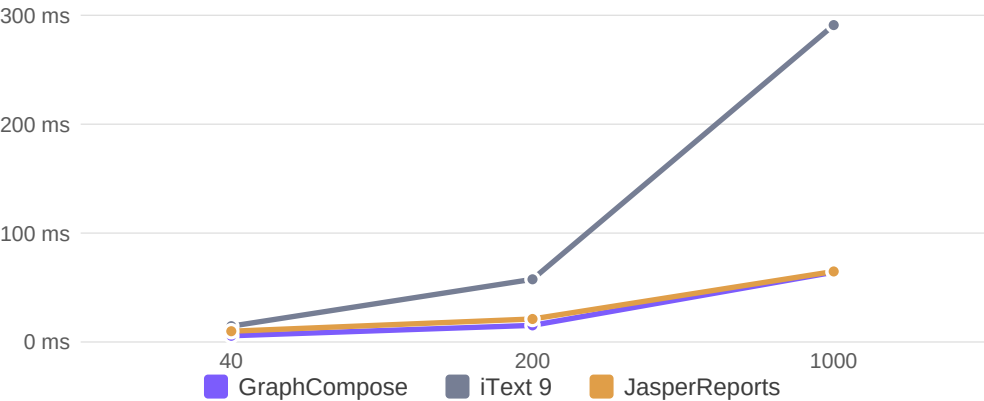


Measured 2026-07-26 23:00:56 · 50 warmup / 100 measurement iterations, single machine — numbers vary by hardware and document. Reproduce: [scripts/run-benchmarks.ps1](#) · see [docs/operations/benchmarks.md](#).

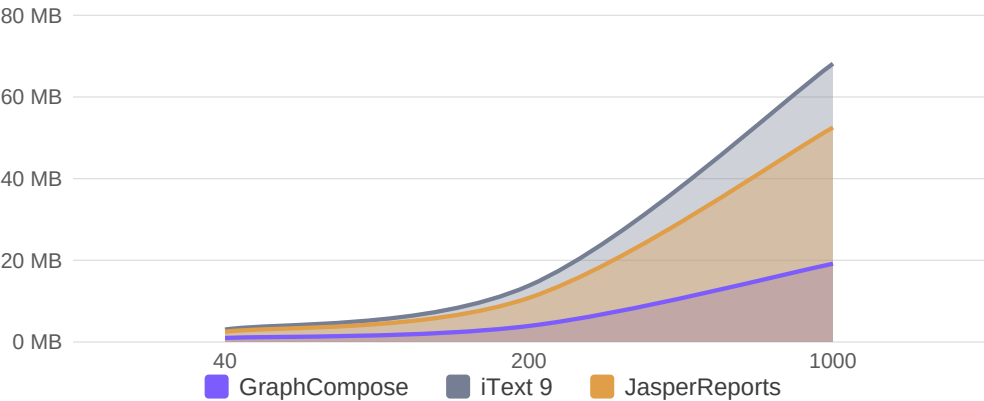
How GraphCompose scales

As the report grows from 40 to 1000 rows, the time lead over iText widens; JasperReports closes to roughly the same render time at 1000 rows, yet GraphCompose stays markedly lighter on memory than both rivals throughout — every series below reads from the same benchmark file.

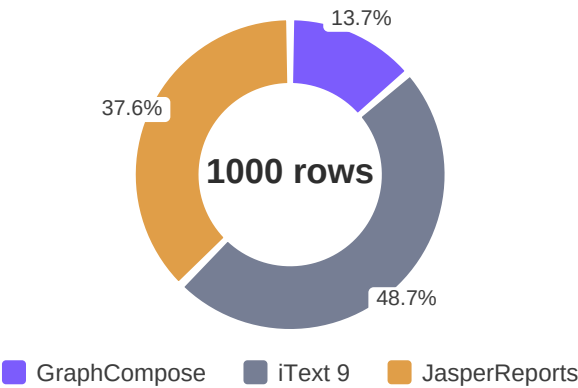
Render time vs. report size — ms



Peak heap vs. report size — MB



Memory at 1000 rows — share of total



Throughput at 1000 rows — docs/sec (higher is better)

